Algorithm 1: FindPart-w-ϵ with a genetic distance constraint and BIC

$U \leftarrow \{\}$; $S_n \leftarrow \{P_{(0,1,\ldots,n-1)}\}$;
$w \leftarrow$ the given integer beam width ($1 \leq w \leq \infty$);
$\epsilon \leftarrow$ the given genetic distance threshold;
$N_{\text{obs}} \leftarrow$ the number of observations;
Conduct maximum likelihood estimation under $M_{(0,1,\ldots,n-1)}$;
$ll_{\text{finest}} \leftarrow$ the maximum log-likelihood of $M_{(0,1,\ldots,n-1)}$;
$BIC_{\text{finest}} \leftarrow (2n-2)\log(N_{\text{obs}}) - 2 \cdot ll_{\text{finest}}$;
$(P_{\text{min}}, BIC_{\text{min}}) \leftarrow (P_{(0,1,\ldots,n-1)}, BIC_{\text{finest}})$;
for $h \leftarrow n-1, n-2, \ldots, 1$ **do**
    $S_h \leftarrow \{\}$; $T_h \leftarrow \{\}$;
    for $P_X \in S_{h+1}$ **do**
        for P_Y such that P_Y is an immediate aggregation of P_X obtained by merging blocks B_s and B_t **do**
            if $\exists v_i \in B_s$ and $v_j \in B_t$ such that $\delta(v_i, v_j) \leq \epsilon$, and P_Y is not an aggregation of any partition in U **then**
                $T_h \leftarrow T_h \cup \{P_Y\}$;
    for $P_Y \in T_h$ **do**
        Conduct maximum likelihood estimation under M_Y;
        $ll_Y \leftarrow$ the maximum log-likelihood of M_Y;
        $BIC_Y \leftarrow (h+n-2)\log(N_{\text{obs}}) - 2 \cdot ll_Y$;
        if $BIC_Y < BIC_{min}$ **then**
            $(P_{\text{min}}, BIC_{\text{min}}) \leftarrow (P_Y, BIC_Y)$;
        $lbBIC_Y \leftarrow BIC_Y - (h-1)\log(N_{\text{obs}})$;
        if $lbBIC_Y > BIC_{min}$ **then**
            $U \leftarrow U \cup \{P_Y\}$;
        else
            $S_h \leftarrow S_h \cup \{P_Y\}$;
    if S_h *is empty* **then**
        break;
    else
        if $w \neq \infty$ **then**
            $S_h \leftarrow$ a set of at most w partitions in S_h achieving the smallest BIC values;
return (P_{min}, BIC_{min});